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Early Onset Prediction for Rotors in Vortex Ring State

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Abstract: The OpenFOAM version 5.0 CFD package was used to investigate detection methods for a small quadrotor drone entering the vortex-ring state. The vortex-ring state causes increased thrust peak-to-mean fluctuations that can be partially attributed to a breakdown of the stable helical wake vortex into a single vortex-ring trapped around the rotor tip. These thrust fluctuations cause significant flight control issues for rotorcraft. Detection of a rotorcraft approaching this flight condition becomes important to avoid any loss of vehicle control. A method for detecting the onset of the vortex-ring state is proposed using various differential pressure measurements on the quadrotor. It has been shown that, by monitoring the pressure difference between the top of the quadrotor body and the bottom of one of the quadrotor legs, a pressure drop can be observed just prior to the onset of the vortex-ring state. This pressure drop was shown to occur during descending flight regimes and may prove to be an effective pre-vortex-ring state warning system. **DOI: 10.1061/(ASCE)AS.1943-5525.0001194.** *© 2020 American Society of Civil Engineers.*

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Introduction

The vortex-ring state is a highly unsteady and turbulent flight regime that a rotor encounters when descending into its downwash. Descending into the downwash of a rotor beyond a certain point causes the stable helical wake vortex structure to collapse into a single strong vortex ring that encompasses the rotor tips (Rodriguez 2007; Rumsey 1987). A reduction in the effective angle of attack of the rotor tip, in combination with an increased inflow velocity into the rotor plane, causes reductions in thrust forces and an increase in thrust force oscillations about a mean thrust value (Brand et al. 2007; Azuma and Obata 1968; Betzina 2001; Mullen and Bernini 2016). The ability to predict the onset of a vortex-ring state remains an important research venture in both the computational and experimental fields of rotor aerodynamics.

Avoiding the vortex-ring state in rotorcraft currently requires either generous safety margins placed around the descent-velocity range of a vehicle or an extremely comprehensive set of experimental data over a range of flight maneuvers to better predict these phenomena (Varnes 1999). Even these results are susceptible to experimental errors or deviations in flight, such as when wind is present. Therefore, being able to predict the onset of the vortex-ring state through an inflight monitoring system would provide a significant benefit to safety for rotorcraft operators and flight vehicles.

The unsteady nature of the vortex-ring state renders the use of classical momentum theory approaches for modeling this state to be ineffective (Taamallah 2010; Leishman 2000; Ribera and Celi 2007; McCormick 1979). In this work, the vortex-ring state is defined using two boundaries—the upper and lower boundary—which is based on the magnitude of the descent rate. The upper boundary is defined as having a smaller magnitude of descent or being closer to the hovering condition, whereas the lower boundary is at a higher magnitude of descent. These boundaries are said to be in the range of hovering (upper boundary) to a descent of twice the rotor-induced velocity at hover (lower boundary), V_h. As the rotorcraft continues to increase its descent rate past the lower vortex-ring state boundary, it enters the post-vortex-ring state flight regime commonly referred to as the windmill brake state, and momentum theory can be reapplied (Taamallah 2010). The experimental work of Wolkovitch (1972) shows that a descent velocity magnitude of $0.707V_h$ is the vortex-ring state upper boundary at which faster descent rates cause the vortex-ring state to begin. The velocity, V_h, is defined as

$$V_h = \sqrt{\frac{T_h}{\rho A}} \tag{1}$$

where T_h = thrust at hover (N); ρ = fluid density (kg/m^3); and A = rotor disc area (m^2).

The vortex-ring state may be identified through an investigation of rotor thrust fluctuations or by flow visualization of the rotor wake breakdown. Thrust trends indicate a significant degradation of thrust performance due to the vortex-ring state and thrust oscillations indicative of the unsteady nature of this flight regime. Thrust oscillations in the range of 12%–20% of the mean thrust coefficient have been displayed both experimentally and numerically (Wolkovitch 1972; Taamallah 2010; Stack et al. 2004; Betzina 2001). Rotor geometry, such as the collective pitch angle, have some effect on the magnitude of the thrust oscillations. Shetty and Selig (2011) indicated that thrust oscillation changes of ±30% in relation to an increased collective pitch angle and a decrease in the freestream advance ratio at which the minimum thrust point occurs (Shetty and Selig 2011). Thrust oscillations may also translate to

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power oscillations if the collective pitch angle of the rotor is high enough (Azuma and Obata 1968). The magnitude at which the vortex-ring state affects the thrust of a rotor and the velocity at which it begins has also been indicated to depend on the rotor blade loading and the spanwise blade twist distribution (Brown et al. 2002). In terms of flow visualization, a breakdown in the stable helical rotor wake vortex may be observed to shift into a single donut-shaped wake vortex that is trapped around the rotor-plane tip. Flow visualization has been used both experimentally through smoke/water tunnels or numerically through Q-criterion and vorticity to identify the vortex-ring state.

Mitigation and avoidance strategies for the vortex-ring state continue to be the main points of interest within the realm of rotorcraft flight. Current measures are limited to avoiding purely descending flights and/or employing alternate flight maneuvers that assist in keeping the shed wake vortex away from the rotor plane during descent. If descent angles are minimized, such as less than or equal to 30° descent angles, the vortex-ring state is nonexistent at virtually all descent speeds (Taamallah 2010). The lack of vortex-ring state effects at low descent angles is because of the forward velocity preventing a wake vortex buildup in the rotor plane. Betzina (2001) supported this claim by reporting that entering the vortex-ring state is not possible at descent angles less than 20°. For this study, the authors used a tiltrotor in helicopter mode in the National Aeronautics and Space Administration's (NASA) Ames wind tunnel (Betzina 2001).

Employing flight maneuvers is currently the optimal way for rotorcraft to avoid or minimize the impact of the vortex-ring state on flight safety. One possible maneuver is to descend into the windmill brake state through a rapid increase in descent speed, which requires sufficient ground clearance. Dziubinski and Stalewski (2007) used flow visualization of their results to indicate that this maneuver effectively pushes the vortex ring above the rotor plane and, consequently, allows for thrust to recover (Dziubinski and Stalewski 2007). Vice versa, a decrease in descent speed is also capable of helping a rotorcraft exit vortex-ring state, as indicated by Grzegorczyk (2013); however, this results in slower thrust recovery due to the strength of the vortex-ring. Spiraled flight patterns or increased forward speed maneuvers are two other approaches that may be used (Taamallah 2010; Chenglong et al. 2015). Mullen and Bernini (2016) used a descent velocity sweep to study the rapid thrust reduction experienced by the rotor on entering the vortex-ring state followed by the steady thrust recovery in the post-vortex-ring state regime (Mullen and Bernini 2016). Ahlin and Brown (2005) demonstrated that rotor thrust recovery begins to occur before the rotor wake stabilizes fully into its windmill brake state structure (Ahlin and Brown 2005).

The extension of vortex-ring state effects on rotor performance has been largely limited to the single rotor and helicopter models, and little investigation has occurred on its impact on multirotor systems (Foster and Hartman 2017). Foster and Hartman (2017) studied the quadrotor's performance under varying descent velocities and descent angles to evaluate the impact on thrust fluctuations, mean thrust, and rotor interference. They concluded that a quadrotor experiences unintended pitch and roll moments potentially due to frequency and phase differences resulting from the individual thrust fluctuation of each rotor. Furthermore, they proposed that these uncontrolled moments may be a result of rotor interactional effects.

The purpose of this study is to demonstrate how pressure differential measurements may be used to predict when a rotorcraft is about to cross the vortex-ring state upper boundary. This work begins with a validation study of a single rotor to ensure that the OpenFOAM computational model is capable of accurately calculating rotor performance versus the experimental data. Next, the same OpenFOAM numerical methodology was applied to a quadrotor drone to investigate the ability of the computational model to capture the relevant vortex-ring state effects. The ability to model the vortex-ring state regime was demonstrated through a descent velocity sweep from hover to a high descent rate. Then, the rotor thrust fluctuation and mean thrust versus descent rate and flow visualizations of the rotor wake were used to indicate that the quadrotor entered into the vortex-ring state. Finally, three pressure ports were defined around the drone fuselage, and pressure differential measurements were calculated to present a means of predicting the onset of the vortex-ring state.

Numerical Methodology

This section presents the computational model formulation used for the single-rotor validation study. The results obtained from the OpenFOAM model are compared with previously obtained experimental data in a low-speed, closed-circuit wind tunnel. A two-mesh grid study is then performed and validated using the Richardson extrapolation. The purpose of this section is to demonstrate the ability of the proposed model to compute realistic rotor performance versus the experimental results.

A numerical model capable of analyzing the unsteady aerodynamics of a quadrotor was formulated in OpenFOAM with a sliding mesh interface to allow for rotor rotation. A Reynolds-averaged Navier–Stokes (RANS) formulation has been indicated to be unsuitable for modeling temporal aerodynamic phenomena based on the work by Westbrook-Netherton and Toomer (2015). Therefore, a detached eddy simulation (DES) approach was used in combination with RANS modeling to capture any transient effects occurring during the vortex-ring state (Yoon et al. 2016; Weerasinghe and Monasor 2017). A DES formulation provides a more computationally efficient approach because RANS modeling is used in the near-wall treatment, whereas large eddy simulation (LES) modeling is used in the freestream region to better capture the wake structure. This methodology helps maintain better usage of computational resources versus a full LES model while providing better accuracy than a purely RANS model.

The presented numerical results are based on an unsteady simulation run for five revolutions for a total of 1,800 timesteps (1 timestep per 1° of rotor rotation). Convergence was reached by the end of the fourth revolution, allowing for the final revolution to be time-averaged. The numerical results were validated using a comparison to the experimental results obtained in the large subsonic wind tunnel of Ryerson University. Details regarding the experimental results can be found in Kolaei et al. (2018). Validation was performed only on the single-rotor model to determine the grid density required to accurately compute the rotor aerodynamic performance. The subsequent quadrotor model was not investigated through a validation study, but it maintains nearly identical grid density and spacing. The only difference between the validation model and the quadrotor model is the physical geometry under investigation.

A one-rotor 46-cm (18-in.) rotor geometry was used as the validation model. This rotor geometry has been labeled the "T-motor rotor" or "T-motor" within the Ryerson Applied Aerodynamics Laboratory of Flight (RAALF) and is referred to as such for the remainder of this manuscript. Fig. 1 plots the T-motor chord and twist distribution from rotor hub to rotor tip, where c = airfoil chord length; R = rotor radius; r = distance from rotor hub to a given point on the rotor blade; and β = rotor pitch angle. More details on the rotor geometry can be found in Kolaei et al. (2018).

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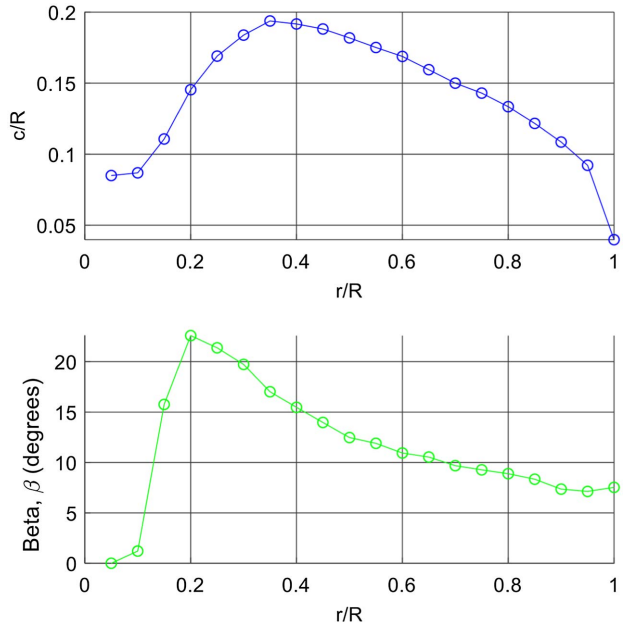


Fig. 1. T-motor rotor chord and twist distribution from hub to tip.

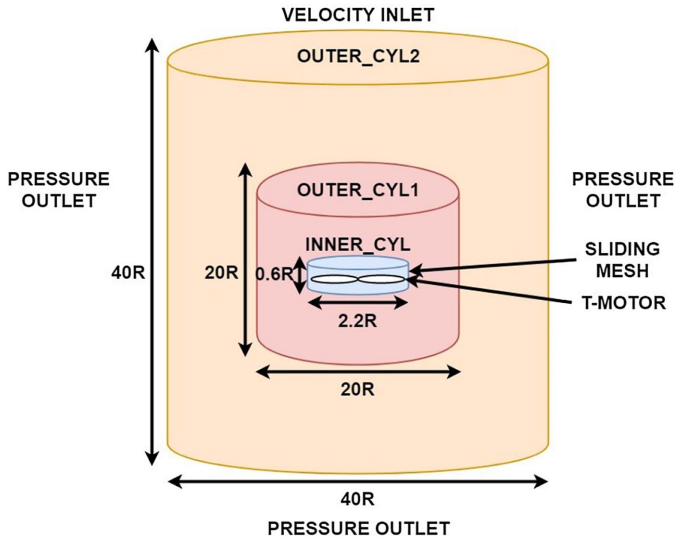


Fig. 2. T-motor computational domain, model sizing, and boundary conditions.

The T-motor computational domain was designed to help control grid spacing and to minimize the boundary condition influence on the flow structure. The control volume used had three subregions, as indicated in Fig. 2, to better control grid spacing and element size transition rates. The body sizing provided in Fig. 2 is defined with respect to the rotor radius, R. Boundary conditions are located 20-blade radii away from the model's origin, or the rotor's center of rotation, which was assumed to be sufficient in limiting the interference between the rotor flow structure and the prescribed boundaries. The location of the external boundaries used was similar to Diaz and Yoon (2018).

OpenFOAM version 5.0 has a library of CFD solvers that may be used for a multitude of applications. The *PimpleDyMFoam* solver is a transient and incompressible flow solver for sliding mesh applications. With a rotor rotation speed of 3,000 rpm, tip speeds

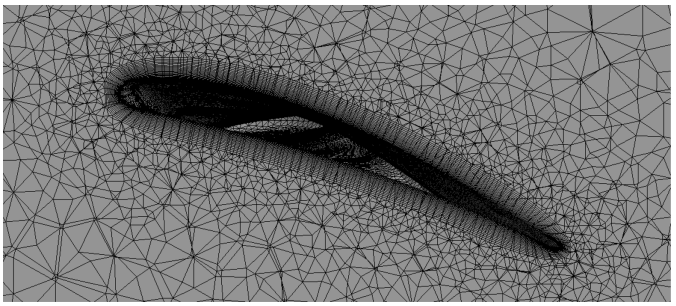


Fig. 3. T-motor boundary layer mesh.

were estimated to be no higher than Mach 0.23, thus rendering the compressibility effects negligible. Model patches surrounding the T-motor were set as arbitrary mesh interfaces (AMI), which allowed the modeling of the rotor rotation and served as an interpolation interface between the rotor and the remainder of the freestream gas path. Boundary patches at the model extremities were set to velocity-inlet and pressure-outlet conditions depending on the need to model climbing or descending cases.

Element sizing and transition rates within the mean flow and in the boundary layer play an important role in the accurate computation of the rotor performance. Element sizing is at its maximum at the outermost subregion and grows steadily more refined toward the rotor itself to better predict viscous interactions within the boundary layer and complex flow structures that are expected to occur during the vortex-ring state. Boundary layer meshing involves 20 layers of structured prism elements that are normal to the wall surface, as indicated in Fig. 3. The total grid refinement of all subregions was 40 million elements.

The two-equation, k-ω shear stress transport (SST) model was implemented for predicting viscous interactions. The turbulent kinetic energy (k_{T_∞}) and turbulent frequency (ω_∞) were adjusted to match the values expected within the wind tunnel at which the experimental results were obtained. Based on hotwire anemometry tests, the turbulent intensity was measured to be approximately 0.25%, as observed in Kolaei et al. (2018). The turbulent frequency and turbulent kinetic energy are functions of turbulent intensity, Tu_∞, freestream velocity (U_∞), kinematic viscosity (ν), and turbulent kinematic viscosity (ν_T)

$$k_{T_\infty} = 1.5(Tu_\infty U_\infty)^2 \tag{2}$$

$$\omega_\infty = C_{\mu,std}\left(\frac{k_{T_\infty}}{\nu}\right)\left(\frac{\nu_T}{\nu}\right)^{-1} \tag{3}$$

where $C_{\mu,std}$ is a model constant used for tuning the turbulent frequency. Based on the work by Pope (2000), a value of 0.09 was used. All other variables were set to match sea-level conditions, and the turbulent viscosity ratio (ν_T/ν) was set to 100 (Pope 2000).

The thrust and power coefficients, C_T and C_P, respectively, are used to present rotor performance. Aerodynamic force coefficients are plotted against the freestream advance ratio, μ, which is a ratio of the freestream velocity to the rotor blade tip speed

$$\mu = \frac{U_\infty}{\Omega R} \tag{4}$$

where Ω = rotor rotational speed (rads/s) and R = rotor radius (m). OpenFOAM presents force calculations in Newtons and Newton-metres for force and moment loadings, respectively, which can be nondimensionalized into thrust and power coefficients (Seckel and Curtiss 1963)

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$$C_T = \frac{2T}{\rho A(\Omega R)^2} \tag{5}$$

$$C_P = \frac{2Q}{\rho AR(\Omega R)^2} \tag{6}$$

where T = thrust force (N); Q = torque on the rotor due to the generated thrust in the direction of the thrust vector (N · m); ρ = air density (kg/m^3); and A = rotor disk area (m^2).

Ryerson University's large subsonic closed-circuit wind tunnel has three antiturbulence screens in the settling chamber upstream of the test section to decrease the incoming airflow's turbulence intensity to approximately 0.25% (Kolaei et al. 2018). The test stand has a fiberglass cowling that encompasses the rotor speed controller, 6-DOF load cell, an electric motor, and other components. Solid blockage corrections are applied to the measured results due to the presence of the test stand, base loads of the test stand, and wall interference effects. Through the Taylor Series Method (TSM) uncertainty analysis, accuracy in thrust and power measurements were estimated to be approximately 1.78% and 2.15%, respectively. A more detailed description of the experimental setup can be found in Kolaei et al. (2018).

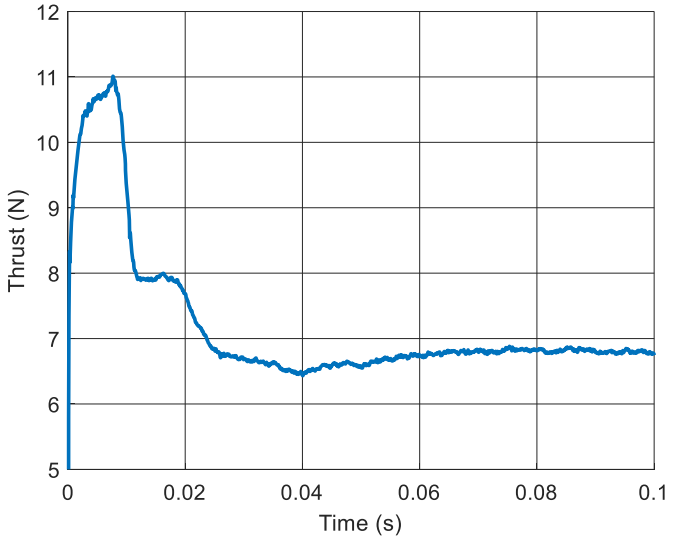


Fig. 4. Convergence history of T-motor rotor at 1-m/s climb rate.

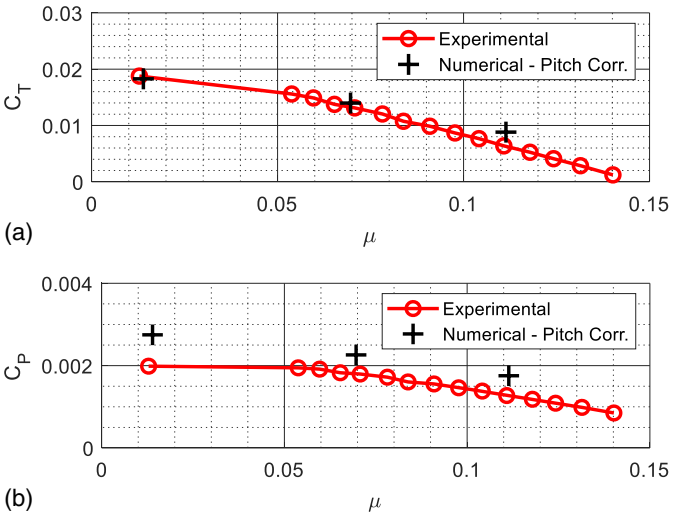


Fig. 5. T-motor numerical and experimental thrust trend. (Data from Kolaei et al. 2018.)

The temporal convergence of the T-motor rotor is provided in Fig. 4. The timestep was defined using the rotor speed, yielding a timestep of 5.6e-5 s (based on a 1 °/timestep rotation rate). Thus, one revolution is equivalent to a simulation time of 0.2 s with the total run equating to 0.1 s or five revolutions. Time-averaging occurs between times of 0.08 and 0.1 s.

The computational results are plotted against the experimental data in Figs. 5(a and b), which indicate the time-averaged thrust and power coefficient trends, respectively. Comparisons are made at three freestream advance ratios: 0.014, 0.070, and 0.111 (1, 5, and 8 m/s). Some errors can be observed between the experimental and numerical data sets that have been attributed to an inaccurate absolute blade pitch angle from the three-dimensional (3D) rotor scan and the inability of an unstructured mesh to effectively capture the rotor leading edge curvature. To correct for difficulties in determining the absolute blade pitch from the 3D scan, the collective pitch of the blade model was adjusted by 4°, which allowed the predicted and measured thrusts to match at near hover conditions within approximately 5%. At this new collective pitch, the predicted power coefficient differed by 24.2% through the experiment.

A grid independence study was also performed to validate the model through a two-mesh study. The first mesh, herein referred to as the "baseline" model, represents the previously mentioned grid used to compare with the experimental data. The second grid, herein referred to as the "refined" model, served as the comparative solution. The grid densities for the baseline and refined models were 40 and 80 million cells, respectively. A Richardson extrapolation was performed between the two grids to estimate the grid convergence index (GCI) value of both grids and the expected error of both models. The results of this study are provided in Table 1. The predicted exact solution at an infinite grid density is a thrust and power coefficient of 0.01690 and 2.54e-3, respectively. The error between the baseline model and the predicted exact solution on the thrust and torque is 7.579% and 7.712%, respectively, which yields a good tradeoff between accuracy and the computational efficiency of running a smaller model.

Quadrotor Drone—Descent Velocity Sweep

This section presents a quadrotor computational model that is used to determine whether the previously defined numerical methodology is capable of capturing vortex-ring state effects. This model maintains identical grid density, as was indicated for the computational baseline model. An axisymmetric, 90° section quadrotor

Table 1. Grid independence study results

Grid	No. of elements	C_T	C_P	Error, C_T (%)	Error, C_P (%)	GCI, C_T	GCI, C_P
Baseline	38,416,666	0.01818	2.735e-3	7.579	7.712	0.2173	0.2210
Refined	80,728,414	0.01768	2.659e-3	4.620	4.701	0.1325	0.1347

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model about the vertical axis was used to simplify the model under investigation. As a result, the interactional effect that would exist between nearby rotors has been neglected due to this formulation; however, this simplification allows for significantly reduced computational requirements being needed for the investigation.

A descent velocity sweep was simulated from a climbing velocity ratio of 0.22 to a 4.30 descent-velocity ratio, with each descent velocity being an independent simulation. The velocity ratio is defined as the ratio of the rotor climb/descent speed to the rotor-induced velocity at hover (Leishman 2000). The vortex-ring state is observed to occur because of rotor thrust fluctuations, rotor wake breakdown, and rotor thrust degradation. Important to note is that the inlet velocity magnitude is in full effect from the start of the simulation, which is representative of a quadrotor instantaneously accelerating to a given descent velocity. Future studies need to simulate smooth transitions from near-hover conditions to a given descent rate for a more accurate representation of the quadrotor flow physics.

Quadrotor Computational Model

A computational model was developed to resemble a typical quadrotor configuration based on geometric files provided by Aeryon Labs Inc. (now part of FLIR systems). A finalized model rendering is provided in Fig. 6. The body has a diameter of approximately 24 cm (9.5 in.), and the quadrotor has been fitted with the Aeryon185 rotor, which is 47 cm (18.5-in.) in diameter. Fig. 7 presents the rotor chord and twist distributions as a function of the distance from the rotor hub.

The computational model was again designed to manage grid spacing in the freestream and boundary layer regions. Fig. 8 provides a schematic for the computational model and presents model sizing based on the rotor radius, R, and the boundary conditions. Fig. 9 presents the same model in an isometric view to better present the model geometry and domain. Model sizing was made relative to the quadrotor centroid. The model used in this section maintains the same grid density to the T-motor model previously validated. As a result, no additional grid independence study was performed on this model. Additional subregions were modeled into the computational schematic to better control grid density and more gradual grid spacing that allows for a more effective detached eddy simulation (DES). The Aeryon185 is mounted to the drone arm, as observed in Fig. 10. The interface between the rotor subdomain and freestream domain is the location of the AMI patch for the sliding mesh interface.

The downwash at hover was determined using momentum theory [Eq. (1)], and the thrust was predicted at the near hover conditions of a 0.22 climbing velocity ratio by the computational model. Boundary conditions were set to replicate sea level conditions, and the rotor speed was 3,000 rpm. A velocity ratio of 0 (hover model) was not investigated due to numerical instability caused by no initial inflowing velocity. With the computed numerical solution, the rotor hover-induced velocity, V_h, can be calculated as 4.65 m/s.



Fig. 6. Quadrotor drone CAD model.

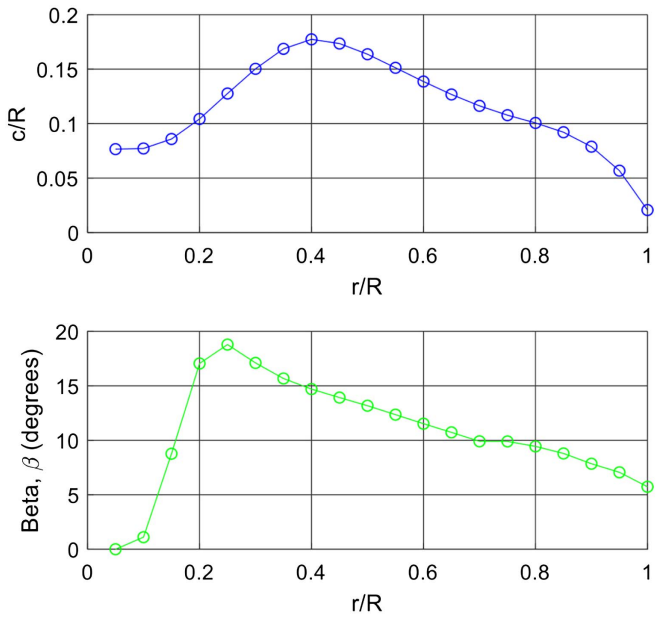


Fig. 7. Aeryon185 rotor chord and twist distributions.

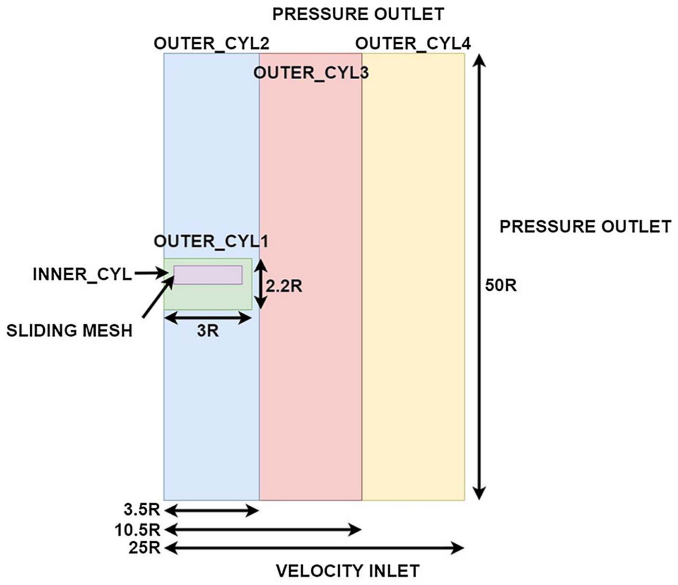


Fig. 8. Computational domain, model sizing, and boundary conditions of quadrotor.

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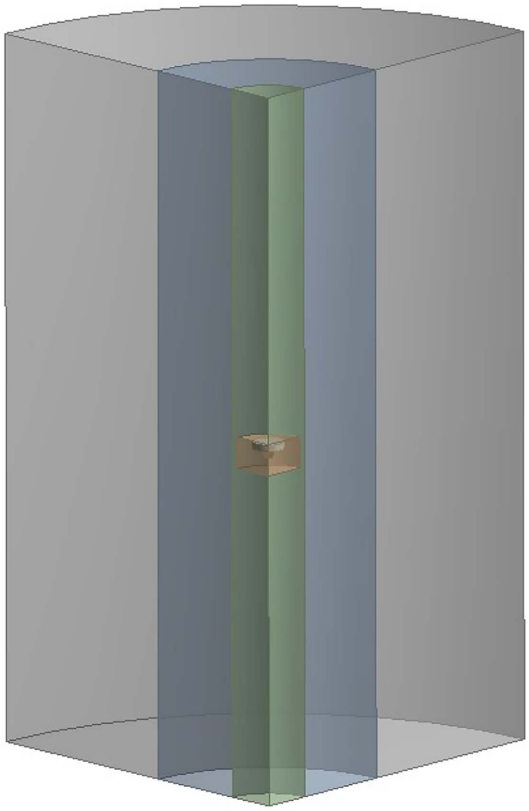


Fig. 9. Computational domain, isometric view.

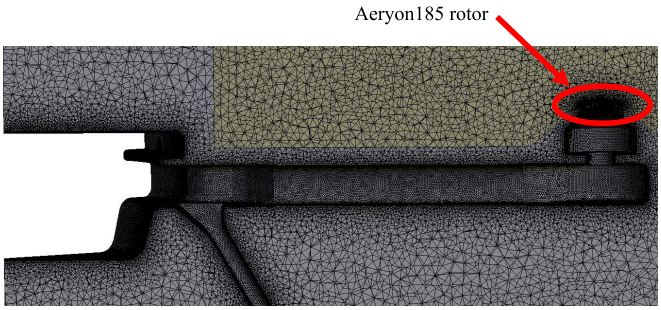


Fig. 10. Quadrotor drone computational grid, 2D section plane.

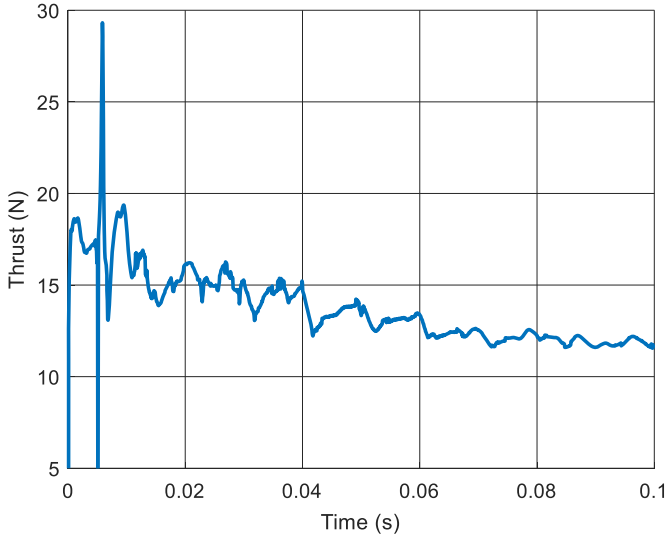


Fig. 11. Aeryon rotor thrust convergence history for 7.5-m/s descent.

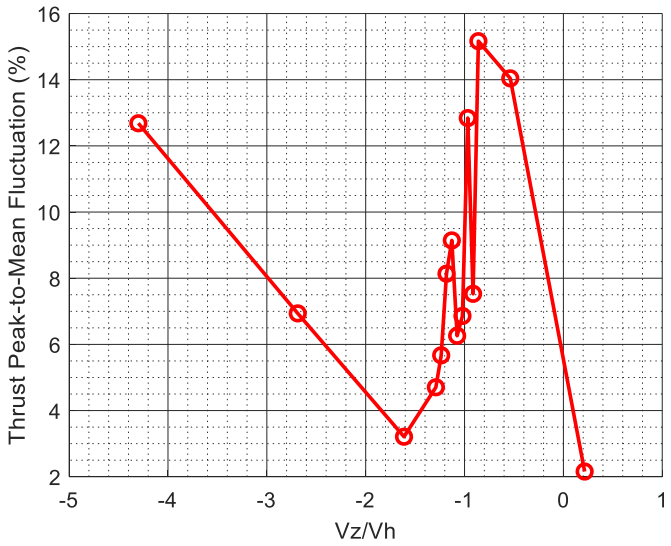


Fig. 12. Rotor thrust fluctuations across descent velocity sweep.

Results

Similar to the validation study, a time-averaged solution can be confirmed by examining the convergence history of the rotor thrust loading, as observed in Fig. 11 for a 7.5 m/s descent rate. The solution is virtually unchanging after a time of 0.08 s, when the time-averaging begins. Any change in the time-averaged thrust solution due to running the case for longer than five revolutions was assumed to be insignificant.

Entry into the vortex-ring state was identified through a number of methods, including thrust fluctuations as a function of the descent rate, flow visualization depicting the rotor wake breakdown, and thrust degradation on entry into the vortex-ring state. Fig. 12 outlines the evolution of rotor thrust fluctuations as the descent rate was increased. Thrust fluctuations were calculated as a percentage of the maximum thrust fluctuation about the time-averaged value for the final revolution. The peak thrust fluctuation was 15.17% and occurred at a descent rate of 4 m/s ($V_Z/V_h = -0.86$). This result aligns with Wolkovitch (1972) as well as a number of other researchers as being the vortex-ring state upper boundary.

Wake breakdown on entry into the vortex-ring state was investigated next through the two-dimensional vorticity contours at different descent rates. The contour plane passes through the rotor hub to show the rotor wake. Furthermore, the contour plot was reflected about the symmetry axis to present an approximation to the flow structure that may be observed by a two-rotor model. However, the reader is reminded that this is only a single-rotor model in combination with the drone fuselage; therefore, any interactional effects between the rotors are not present in the flow contour. Fig. 13 depicts a flow structure around the quadrotor at a 4.5 m/s descent rate ($V_Z/V_h = -0.98$) that lacks any coherent flow structure representing the wake breakdown characteristic of the vortex-ring state. In contrast, at a slightly higher descent rate of 5 m/s ($V_Z/V_h = -1.08$), Fig. 14 clearly indicates how the wake structure has restabilized into a single vortex ring concentrated around the rotor tips.

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The vortex-ring state flight regime was finally identified using the thrust profile with respect to the descent-velocity ratio. Figs. 15(a and b) plot the descent velocity sweep results for the thrust coefficient, C_T, versus the velocity ratio, V_Z/V_h, which is the climbing/descending velocity normalized with the rotor-induced velocity at hover. Rapid degradation in the rotor thrust marks the beginning of the vortex-ring state that, as expected, occurred close to the rotor-induced velocity at hover ($V_Z/V_h = 1$). Maximum thrust loss due to the vortex-ring state occurs at a descent rate of 4.5 m/s ($V_z/V_h = -0.97$), which marks the beginning of this phenomenon. These results imply that the vortex-ring state occurs at 4.5 m/s and faster descent rates and is nonexistent at slower descent rates. The numerical model appears to detect a marginal change in the power coefficient across the vortex-ring state flight regime. The authors can conclude that the vortex-ring state flight regime of the current configuration occurs from a V_Z/V_h of -0.8 to -1.1.

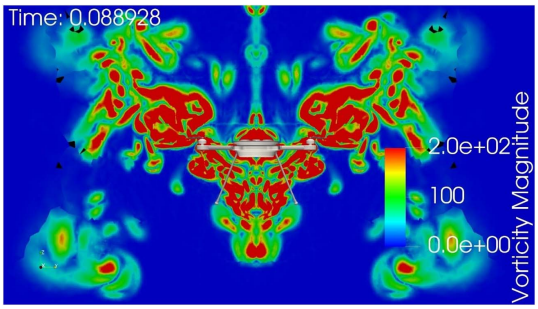


Fig. 13. Vorticity contour around quadrotor at 4.5-m/s descent rate, $V_Z/V_h = -0.98$.

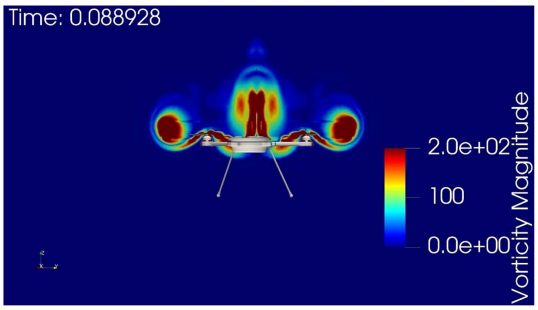


Fig. 14. Vorticity contour around quadrotor at 5-m/s descent rate, $V_Z/V_h = -1.08$.

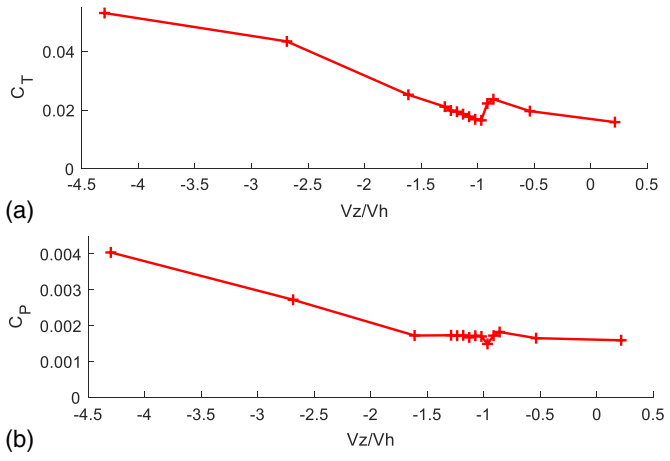


Fig. 15. (a) Quadrotor drone descent velocity sweep, rotor thrust performance; and (b) quadrotor drone descent velocity sweep, rotor power performance.

Vortex-Ring State Prediction by Pressure Differential Analysis

The ability to predict the early onset of the vortex-ring state becomes critical for the safe operation of rotorcraft when descending. One proposed method is to investigate pressure differentials over the quadrotor fuselage to locate a warning sign for quadrotors that are descending too close to the critical descent rate. Three differential pressures were investigated in this study. Three pressure-port locations are defined to assess the three differential pressures. The locations for the pressure-ports can be observed in Fig. 16. Port 1 is located at the top of the quadrotor fuselage and is aligned with a vertical axis that passes through the body centroid. Port 1 is approximately 5-cm above the fuselage surface, conceivably mounted on a sting. Port 2 is located 5-cm below the quadrotor fuselage, also in alignment with the same vertical axis through the body's centroid. Port 3 is located just below one of the legs of the quadrotor landing gear. Important to note is that although all three ports are indicated on the same plane in Fig. 16, port 3 is offset from this plane by approximately 10 cm, directed into the page. The notation for the measured pressure difference, ΔP_{ij}, between point i and j is

$$\Delta P_{ij} = P_i - P_j \tag{7}$$

The effectiveness of differential pressure monitoring can be shown on a plot of pressure versus descent rate, as in Fig. 17. The time-averaged results show the pressure differentials ΔP_{12}, ΔP_{13}, and ΔP_{23} across the previously tested descent velocity sweep. The ΔP_{12} pressure differential picks up an approximate 40-Pa pressure drop when in the vortex-ring state; however, minimal pressure shifts are observed before the vortex-ring state takes effect. This pressure drop is problematic because the indicative pressure drop only occurs once the aircraft has already entered the vortex-ring state regime, and it may already be too late for preventive measures. Alternatively, ΔP_{13} indicates a more significant pressure drop that occurs at descent velocities before the vortex-ring state begins and continues decreasing until just after the vortex-ring state ends. This pressure differential presents a potentially viable option for predicting the vortex-ring state in a purely descending flight maneuver. Similarly, the ΔP_{23} also shows an indicative pressure drop prior to the onset of the vortex-ring state; however, the trend is marred with a subsequent increase in pressure immediately after which may make the pressure differential more difficult to monitor inflight, whereas the ΔP_{13} measurement continues to decrease.

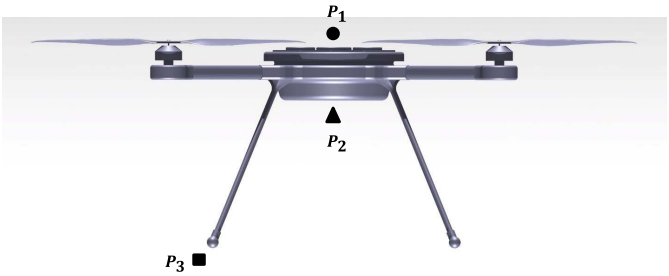


Fig. 16. Quadrotor model with pressure port locations.

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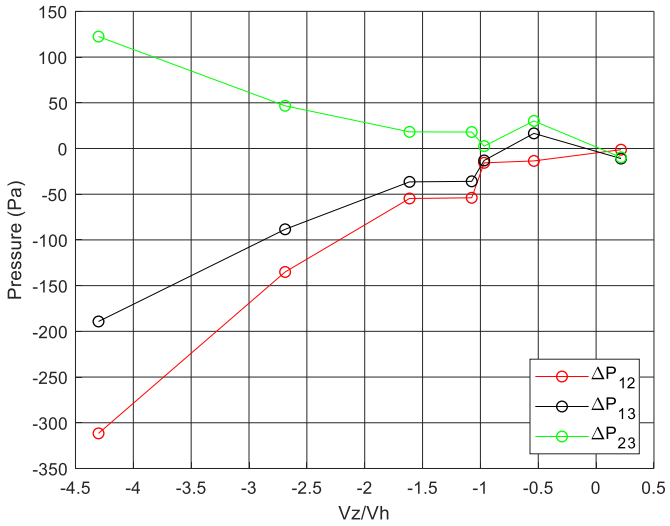


Fig. 17. Pressure differential analysis for early onset vortex-ring state warning system.

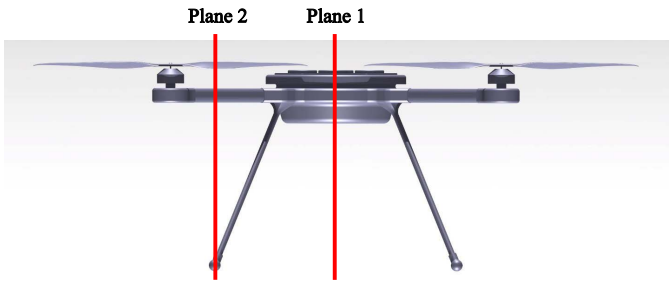


Fig. 18. Quadrotor model, pressure contour plane locations.

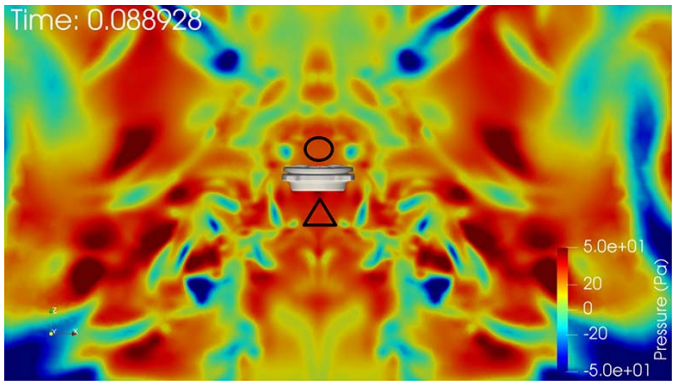


Fig. 19. Pressure contour at Plane 1, $V_z/V_h = -0.538$.

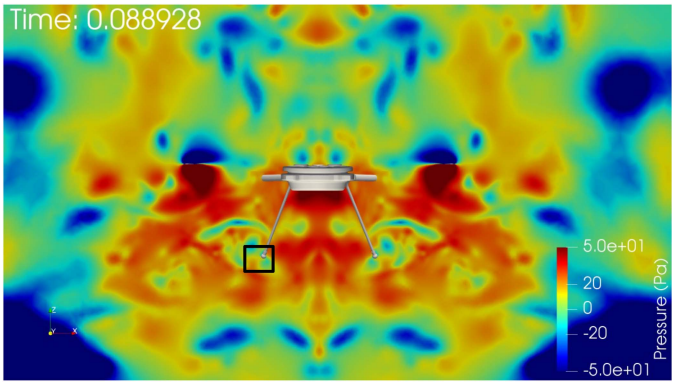


Fig. 20. Pressure contour at Plane 2, $V_z/V_h = -0.538$.

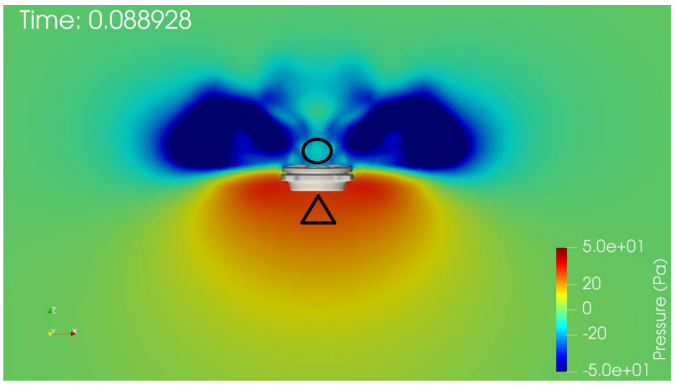


Fig. 21. Pressure contour at Plane 1, $V_z/V_h = -0.968$.

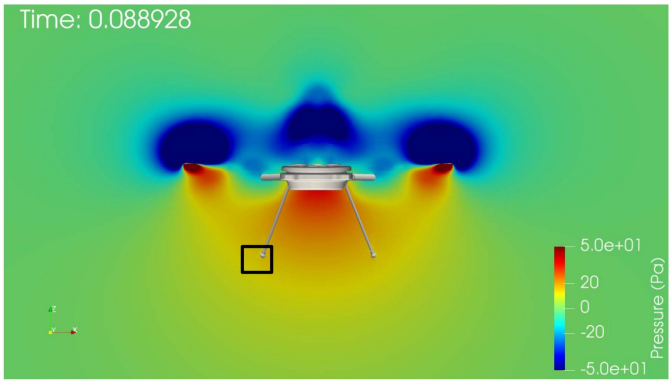


Fig. 22. Pressure contour at Plane 2, $V_z/V_h = -0.968$.

The flow conditions of the quadrotor in the vortex-ring state were visualized using contoured planes at two different cross-sectional locations that are defined in Fig. 18 as Plane 1 and Plane 2. Plane 1 intersects the quadrotor fuselage body where the symmetry boundary condition is located, and Plane 2 intersects through the middle of the end of the quadrotor legs. Pressure contours at these locations are indicated for a descent rate of $V_Z/V_h = -0.538$ in Figs. 19 and 20 and $V_Z/V_h = -0.968$ in Figs. 21 and 22. In these figures, the locations of the pressure ports have been marked using a "circle" for P_1, a "triangle" for P_2, and a "rectangle" for P_3. The contours indicate that the pressures at ports 1 and 2 are affected by the shift into the vortex-ring state flight regime due to their proximity to the rotor. Comparable pressure values are observed. Consequently, both experience a very similar pressure drop as the vortex-ring state boundary is penetrated; therefore, the differential between them is difficult to pick up on. The differential pressure between ports 1 and 3 is more effective because port 1 experiences a large pressure drop as the quadrotor enters the vortex-ring state, whereas port 3 is sufficiently far away from the rotor that it is less impacted. Ports 1 and 3 both experience a pressure drop when entering the vortex-ring state; however, they experience it at different magnitudes. The different magnitudes of the pressure drops allow the ΔP_{13} differential

pressure to predict the onset of the vortex-ring. Further testing along different flight plans is required to determine whether this pressure measurement is unanimously effective in all other flight conditions.

Conclusion

A prediction strategy for the onset of the vortex-ring state was proposed through pressure differential measurements surrounding a computational quadrotor model. The computational model was validated using a single rotor during climbing flight relative to the experimental data obtained in a subsonic wind tunnel and through a grid independence study. The quadrotor numerical model then employed an identical grid density to the validation model and was tested at various descent velocities to examine the computational model's ability to calculate the vortex-ring state regime. Finally, a prediction method was proposed by defining various pressure ports around the quadrotor fuselage. A pressure differential probe between the top and bottom of the quadrotor body provided no indication of the onset of the vortex-ring state until the quadrotor had already entered this flight regime. However, a differential pressure probe between the top of the drone body and the bottom of one of the quadrotor legs indicates a pressure drop well before the vortex-ring state takes effect. This result indicates that an early warning system of the vortex-ring state for quadrotors may be possible by measuring pressure differentials in this setup. Continued study of different flight paths must also be performed to determine whether the proposed system is suitable for other flight conditions. Furthermore, the implemented system needs to recognize the difference between a pressure drop due to changes in airflow conditions versus a pressure drop caused by the onset of the vortex-ring state. This distinction could be made by correlating a measured pressure drop with the vertical velocity component of the quadrotor such that the vortex-ring state is predicted only if the drone descends in combination with a measured pressure drop.

Data Availability Statement

Some of the data, all of the codes, and all of the models used in this study are available from the corresponding author on request.

Acknowledgments

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Notation

The following symbols are used in this paper:

A = rotor disk area;
c = rotor chord;
C_P = power coefficient;
C_T = thrust coefficient;
$C_{\mu,std}$ = model constant;
k_{T_∞} = turbulent frequency;
P_i = measured static pressure at port i;
Q = torque;
r = distance from rotor hub to given point on rotor;
R = rotor radius;
T = thrust;
Tu_∞ = freestream turbulent intensity;
U_∞ = freestream velocity;
V_h = induced velocity at hover;
V_z = rotor climb/descent velocity;
ν = freestream kinematic viscosity;
ν_T = turbulent kinematic viscosity;
β = rotor pitch angle;
ΔP_{ij} = pressure difference between probe i and probe j;
μ_∞ = freestream advance ratio;
ρ = density of air;
Ω = rotor rotation speed; and
ω_∞ = freestream turbulent kinetic energy.

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